

Homework 1

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Problem 1: [20 marks] You run a company called *Mathemagics* which sells textbooks to students. You have determined that:

- At a price of \$100 per textbook, the monthly demand is 200 textbooks.
- At a price of \$80 per textbook, the monthly demand is 240 textbooks.
- The price and the demand are linearly related.
- The variable cost of producing a textbook is \$30.
- There is a fixed cost of \$600 for running the factory which produces the textbooks.

1. Find the linear demand equation relating the price p and the demand q .
 2. Write an expression for the profit of the company as a function of the demand (the price should not appear as a variable).
 3. Suppose that the company is not for profit and just aims to break even so that it does not lose any money. What are the possible values of q for which this holds? Which of these values makes more sense if the company wants to sell as many textbooks as possible?
 4. Assume now that the company owners want to get as rich as possible. Find how many textbooks should be produced per month to maximize the profit. What would the total profit be in this case?
- Hint: If $f(x) = ax^2 + bx + c$ and $a < 0$ the maximum of $f(x)$ is attained at $x = \frac{-b}{2a}$.

Problem 2: [20 points] For each of the following limits indicate either its value or that it does not exist. You must justify your answer. The use of L'Hôpital's rule is not accepted as an argument (although you may use it to check your answer). Note that for solving some of these limits you may need to use some tools which we will learn on the lecture of Friday 14.

1. $\lim_{x \rightarrow 2} e^x + x^2 - 5$
2. $\lim_{x \rightarrow 1} \frac{x^{9000} - 1}{x - 1}$
3. $\lim_{x \rightarrow 0} \frac{\sqrt{x+1}}{x}$
4. $\lim_{x \rightarrow 0} \frac{\sqrt{1-x} - \sqrt{1+x}}{x}$
5. $\lim_{x \rightarrow 0} \cos\left(\frac{1}{x^2}\right)$

Problem 3: [10 points] Give an example of a function $f(x)$ for which the limit from the left is zero, but the limit from the right does not exist, that is:

$$\lim_{x \rightarrow 0^-} f(x) = 0 \text{ but } \lim_{x \rightarrow 0^+} f(x) \text{ does not exist}$$

Updated September 12, 2018.